# AdaptConfig

AI-Powered Integration Configuration Platform
for Enterprise Lending

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https://adaptconfig-frontend-production.up.railway.app

# The Problem

- Credit Bureaus (CIBIL)
- Aadhaar eKYC
- GST Verification
- Payment Gateways
- Fraud Detection
- SMS / Email / AA

**Pain Points**

- **2–4 weeks** per integration
- Field mapping errors → data loss
- No pre-production testing
- 8 APIs = **16–32 weeks total**

Cost: 8 APIs × Rs 1–2L each = **Rs 8–16 Lakhs**

# Our Solution

**Upload API Spec** $\xrightarrow{\text{AI}}$ **Auto-Config** $\xrightarrow{\text{Test}}$ **Deploy**

From **weeks** to **minutes**. Zero manual field mapping.

| Metric | Result | |
|---|---|---|
| Endpoints parsed | 4 | from CIBIL v2 spec |
| Fields extracted | 28 | with types & required flags |
| Auth schemes | 2 | OAuth2 + API Key |
| Parse confidence | 95% | |
| Fields mapped | 28/28 | at 100% confidence |
| Simulation | 8/8 PASSED | all steps green |
| Total time | ¡ 5 seconds | end to end |

# Architecture

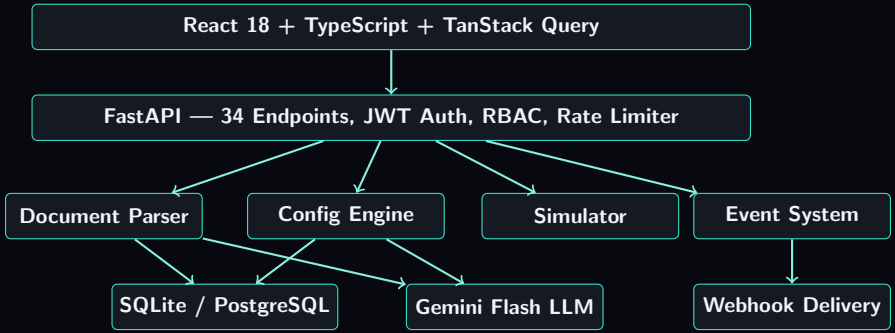
**React Frontend** $\xrightarrow{\text{JWT}}$ **FastAPI Backend (34 endpoints)** $\rightarrow$ **PostgreSQL**

| Layer | Technologies |
|---|---|
| Frontend | React 18 · TypeScript · Tailwind CSS · Recharts |
| Backend | FastAPI · SQLAlchemy (async) · Pydantic v2 |
| AI Engine | Google Gemini 3 Flash (REST API) |
| Services | Document Parser · Config Engine · Simulator |
| Database | PostgreSQL (prod) · SQLite (dev) |
| Deploy | Railway · Docker · nginx · Alembic migrations |
| Auth | JWT · PBKDF2 · RBAC · per-user tenant isolation |

Backend serves 34 API endpoints across 10 route modules.

AI pipeline: Gemini 3 generates config $\rightarrow$ Rule engine validates with fuzzy matching.

# Architecture Deep-Dive



# Config Generation Pipeline

**Hybrid AI + Rule Engine:**

1. Gemini Flash generates config from document + adapter schema
2. **Rule engine** augments with:
   - 100+ Indian fintech synonyms
   - Fuzzy string matching (rapidfuzz)
   - Jaccard token overlap
3. Rule-based confidence **overrides** LLM
4. Unmapped fields **backfilled**

**Graceful degradation:**

| Path | When |
|---|---|
| LLM + Rules | API key present |
| Rule fallback | LLM fails |
| Pure rules | AI disabled |

**Field matching tiers:**

1. Exact synonym → 1.0
2. Fuzzy match → 0.6–0.99
3. Token Jaccard → 0.6+

# Rollback & Version Control

**Configuration Lifecycle:**

$$\text{Draft} \rightarrow \text{Configured} \rightarrow \text{Validating} \rightarrow \text{Testing} \rightarrow \text{Active}$$

**Version history:**

- Every change creates a snapshot
- Full state serialization
- One-click rollback to any version
- Pre-rollback snapshot for safety

**Diff engine:**

- Recursive dict/list comparison
- Breaking change detection
- Side-by-side version comparison
- Export as JSON or YAML

**Audit trail:**

- Immutable log of every action
- Actor, timestamp, resource details
- Per-user data isolation

# Database Schema (11 Tables)

| Table | Purpose | Key Relations |
|---|---|---|
| `users` | Auth, RBAC (admin/editor/viewer) | tenant owner |
| `documents` | Uploaded specs (PDF/DOCX/YAML) | tenant-scoped |
| `adapters` | 9 pre-built adapters | global catalog |
| `adapter_versions` | Multi-version per adapter | FK → adapters |
| `configurations` | Generated integration configs | FK → adapter, doc |
| `config_history` | Versioned snapshots for rollback | FK → configs |
| `simulations` | Test run results | FK → configs |
| `simulation_steps` | Per-step test details | FK → simulations |
| `audit_logs` | Immutable action log | tenant-scoped |
| `webhooks` | Registered callback URLs | tenant-scoped |
| `webhook_deliveries` | Delivery attempts & status | FK → webhooks |

UUID primary keys · Row-level tenant isolation · Alembic migrations · CASCADE deletes

# Authentication & User Isolation

**Security:**

- JWT access (30min) + refresh (7d) tokens
- PBKDF2-SHA256 password hashing with salt
- Per-user data isolation (unique tenant ID)
- Role-based access: admin / editor / viewer

**Each user sees only their own:**

Documents, Configs, Simulations, Audit

Adapters are shared (global catalog).

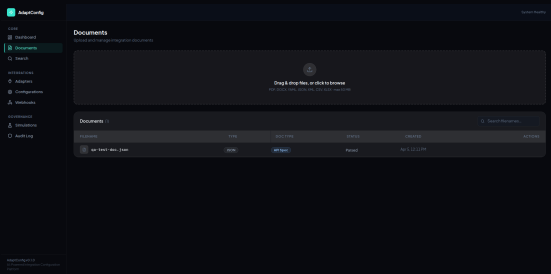
**Additional protections:**

- CORS restricted to known origins
- CSP, HSTS, X-Frame headers
- Rate limiting (100 req/min)
- PII masking (Aadhaar, PAN, phone)
- AES-256 Fernet encryption
- Path traversal protection
- Alembic database migrations
- Structured JSON logging

# Dashboard

AdaptConfig
System Healthy
CORE
Dashboard
Documents
Search
INTEGRATIONS
Adapters
Configurations
Webhooks
GOVERNANCE
Simulations
Audit Log
S Akash
Sign out
Configuration Summary
1 total configuration
BY STATUS
Configured
1
AVG. CONFIDENCE
64%
across all mapped configurations
Weekly Activity
Documents & configurations processed
3
2.25
1.5
0.75
0
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Documents
Configs
Adapter Status
Connected adapters
Active
8
Data Throughput
Records processed per hour
4 processed
2
1.5
1
0.5
0
00:00
04:00
08:00
12:00
16:00
20:00

# Step 1: Upload & Parse




**Formats:** PDF, DOCX, YAML, JSON

**Parser extracts:**

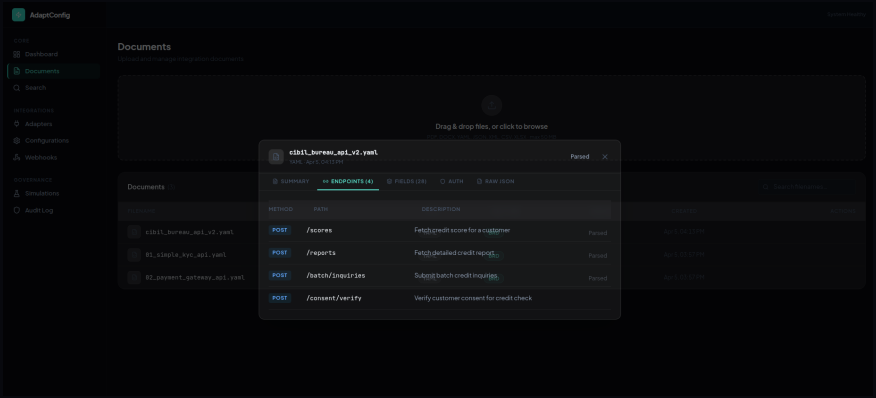
- API endpoints & methods
- Request/response fields
- Auth schemes
- SLA requirements

**CIBIL v2 result:**

4 endpoints · 28 fields · 2 auth

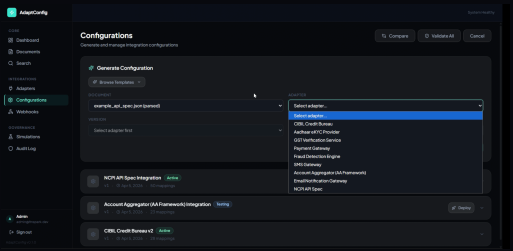
95% confidence

# Parsed Document — 5-Tab Detail View




Summary · Endpoints · Fields (28) · Auth · Raw JSON

# Step 2: AI Config Generation



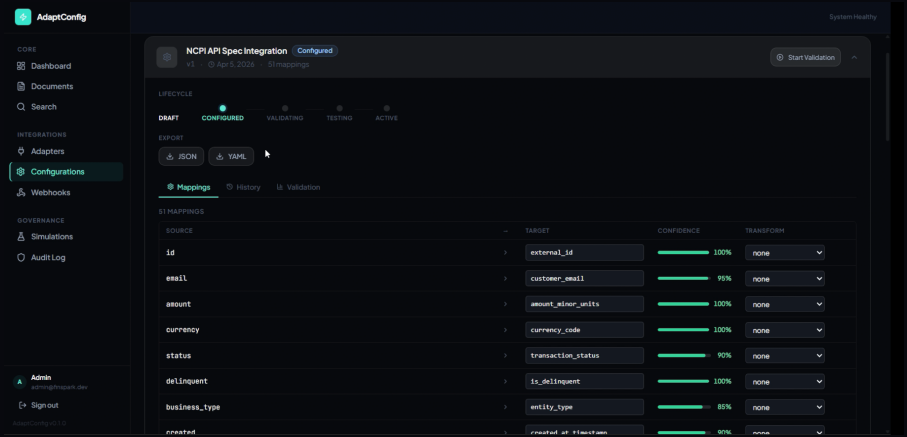

**Two-stage pipeline:**

1. Gemini 3 Flash generates config from document + adapter schema
2. **Rule engine** augments with fuzzy matching, Indian fintech synonyms, confidence scores

**28/28 fields mapped at 100%**

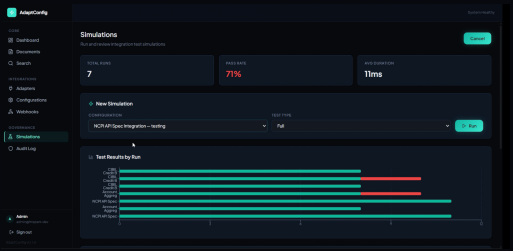
If no adapter matches, auto-detects and prompts to create one from the spec.

# Editable Field Mappings




Lifecycle stepper · Editable targets · Transform dropdowns · Confidence bars · Export · History

# Step 3: Simulation

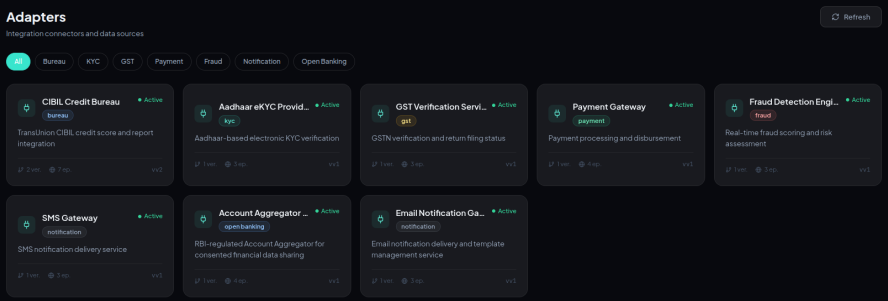



**8-step pipeline:**

- ✓ Config structure
- ✓ Field mapping coverage
- ✓ Endpoint: /scores
- ✓ Endpoint: /reports
- ✓ Endpoint: /batch/inquiries
- ✓ Endpoint: /consent/verify
- ✓ Auth validation
- ✓ Hooks validation

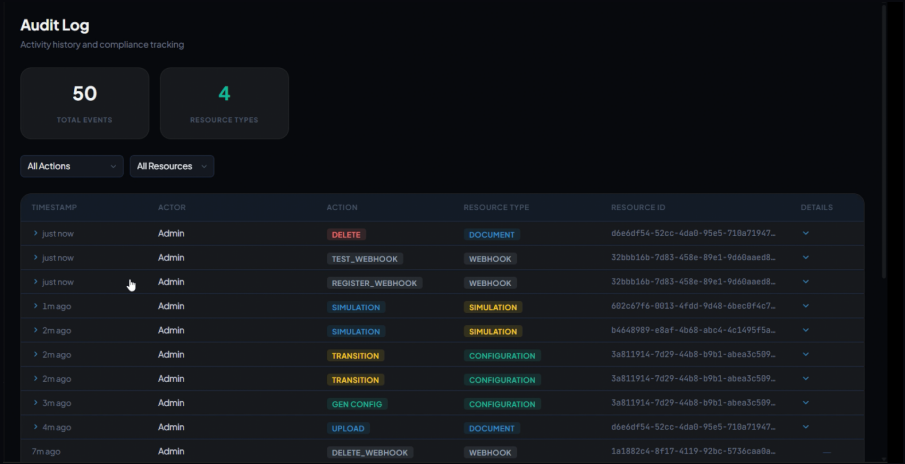
**8/8 PASSED**

# Adapter Catalog (9 Pre-Built + Custom)



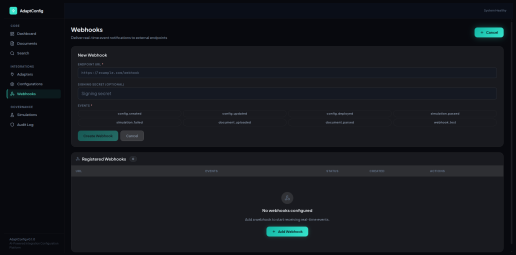

CIBIL · eKYC · GST · Payment · Fraud · SMS · Account Aggregator · Email · Custom

# Audit Trail




Immutable log of every action · Filter by action type & resource · Pagination · Per-user isolation

# Webhooks & Event System




**Event-driven notifications:**

- Register webhook URLs for real-time callbacks
- Subscribe to events: config.created, simulation.completed, document.uploaded
- Test delivery with one click
- HMAC signature verification
- Delivery status tracking

**Use cases:**

- CI/CD pipeline triggers
- Slack/Teams notifications

# Results

| | Manual | AdaptConfig | Improvement |
|---|---|---|---|
| Time per integration | 2–4 weeks | $<$ 5 minutes | $\sim$500x faster |
| Field mapping errors | Common | AI-validated | Near zero |
| Pre-deploy testing | Rare | 8-step pipeline | Always |
| Cost per platform | Rs 8–16L | Minimal | $>$90% savings |

| Metric | Value |
|---|---|
| API Endpoints | 34 |
| Adapters | 9 |
| Tests | 899 |
| Coverage | 82% |
| Pages | 8 |

| Performance | Value |
|---|---|
| Parse confidence | 95% |
| Mapping accuracy | 100% |
| Simulation | 8/8 pass |
| Config gen | $<$ 5s |
| Doc parse | $<$ 1s |

# Live Demo

https://adaptconfig-frontend-production.up.railway.app

1. **Register / Login**
   Create account or use demo credentials
2. **Upload API Spec**
   Drag & drop YAML, PDF, or DOCX
3. **View Parsed Fields**
   28 fields, 4 endpoints, 2 auth schemes
4. **Generate Config**
   Gemini 3 + Rule Engine pipeline
5. **Review Mappings**
   Edit targets, transforms, confidence bars
6. **Simulate & Test**
   8/8 steps pass with mock API responses
7. **Deploy**
   Configured → Validating → Testing → Active

Default admin: admin@finspark.dev / Admin1234!

# Thank You

## AdaptConfig

AI-Powered Integration Configuration Platform

| | |
|---|---|
| Live | `https://adaptconfig-frontend-production.up.railway.app` |
| API | `https://adaptconfig-api-production.up.railway.app/docs` |
| Code | `https://github.com/Akasxh/adaptconfig` |

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# LLM-Powered NLP Architecture

**4 services enhanced with Gemini 2.5 Flash:**

1. Document Parser – LLM extracts endpoints, fields, auth, SLA from *any* doc type (not just BRD/SOW)
2. Field Mapper – Semantic matching replaces fuzzy string + synonym dicts
3. Simulation Validator – LLM analyzes config quality with explanations
4. Search Service – NLU query understanding replaces keyword dicts

**Design principles:**

- Every LLM call has a **rule-based fallback**
- Gated by `FINSPARK_AI_ENABLED` flag
- Zero LLM cost when AI is disabled
- JSON structured output via `responseMimeType`
- Temperature 0.1 for deterministic results
- All calls augmented with regex-detected entities

# LLM Cost Analysis – Gemini 2.5 Flash

| LLM Call | Input (tok) | Output (tok) | Cost/call |
|---|---|---|---|
| Document Parsing | 4,000 | 2,000 | $0.008 |
| Config Generation | 3,000 | 2,000 | $0.008 |
| Field Mapping | 1,000 | 1,000 | $0.004 |
| Simulation Validation | 2,000 | 2,000 | $0.008 |
| Search Query | 500 | 200 | $0.001 |
| **Full workflow** | **10,500** | **7,200** | **$0.028** |

**Monthly cost @ thinking ON:**

100 users: $39/mo ($0.39/user)

1,000 users: $394/mo ($0.39/user)

**With thinking OFF (-78%):**

100 users: $8.50/mo ($0.09/user)

1,000 users: $85/mo ($0.09/user)

Pricing: Gemini 2.5 Flash – $0.30/M in, $3.50/M out (thinking ON) | $0.15/M in, $0.60/M out (thinking OFF)